Barriers to Mobile Banking and Digital Wallet Adoption Among University and College Students in Pakistan

Okasha Saeed
Department of Financial Technology
FAST NUCES Islamabad
Islamabad, Pakistan
i235556@isb.nu.edu.pk

***Abstract*—This paper presents the evaluation of a machine-learning classification system designed to identify the main barriers stopping students in Pakistan from adopting mobile banking and digital wallet services. Using a simulated survey dataset of 200 student responses, two classification models namely Logistic Regression (LR) and Decision Tree (DT)—were trained and used. Barriers were categorised into four classes: Technological, Psychological, Financial, and Social. Results show that Logistic Regression achieved the highest accuracy of 85.0 %, outperforming Decision Tree (60.0 %) and the baseline (20.0 %). The results show that machine learning approaches can classify student barriers and support targeted interventions.**

***Index Terms*—mobile banking, digital wallet adoption, Pakistan, machine learning, classification, barriers, financial inclusion, TAM, UTAUT**

I. Introduction

Mobile banking applications and digital wallets have fundamentally transformed the delivery of financial services across the globe. In developing economies such as Pakistan, this transformation has been particularly significant, offering millions of citizens access to financial tools that were previously unavailable or inaccessible to them. Over the last five years, applications such as JazzCash, Easypaisa, and SadaPay have expanded widely across the country, providing users with access to financial services without the need of a traditional bank account [1]. The growth of these platforms has been incredible, with JazzCash alone reporting tens of millions of registered users and billions of rupees in monthly transactions. These platforms offer services ranging from P2P money transfers and utility bill payments to business payments and microfinance products, making them particularly relevant for young, mobile users such as university students.

Despite this rapid expansion, a large segment of university and collegegoing students in Pakistan continues to hesitate while adopting these tools. The widespread availability of mobile banking applications has not translated into widespread usage among the student population. A significant proportion of students still rely on cash for everyday transactions even when digital alternatives are easily accessible, affordable, and actively promoted by both the government and private sector. Everybody nowadays takes online payments, from street vendors to large business and corporations. [3]. The coexistence of available digital financial tools and persistent cash dependency represents a critical challenge for financial inclusion efforts for Pakistan.

The underlying reasons for this behaviour remain incompletely understood. Factors such as perceived ease of use, security concerns including fear of hacking or fraud, unreliable internet access, limited financial literacy, low trust in digital systems, and social norms may all influence a student's decision to adopt or not adopt mobile banking applications [2]. However, the relative importance of these factors, and the way in which they interact with one another in the specific context of Pakistani university students, has not been properly studied. Without a clear understanding of these dynamics, interventions designed to promote adoption are likely to be generic and ineffective.

This problem carries particular significance given Pakistan's demographic . Pakistan is one of the youngest countries in the world, with a large and growing population of university age citizens who are technologically literate and smartphone equipped. These students represent an important demographic for financial inclusion efforts, as their early adoption of mobile banking habits can have lasting and far reaching effects on broader economic participation and digital activity across society [4]. Research has consistently shown that once young users form digital payment habits, those habits tend to persist into working life and drive wider adoption across households and communities. Targeting this demographic is therefore not only practically important for fintech companies/government seeking to grow their users base, but also very important for national financial inclusion targets.

Existing research on mobile banking adoption has mainly focused on general adult populations, working professionals, or banking customers as groups. Frameworks such as the Technology Acceptance Model (TAM) [9] and the Unified Theory of Acceptance and Use of Technology (UTAUT) [6] have been extensively applied in countries including India, China, Malaysia, Indonesia, and several Western nations. However, these frameworks have rarely been applied in a student specific Pakistani context, and the findings from other countries cannot be assumed to generalise and apply to Pakistani students given

the significant differences in infrastructure, financial literacy levels, cultural norms, and economic conditions [11].

So there is therefore a clear and important gap in the literature, targeting university and college students in Pakistan as a specific group with unique behaviour patterns, technological literacy levels, trust levels, and social influences. Understanding the barriers specific to this group is important and vital for designing effective, targeted financial inclusion strategies. Fintech companies, educational institutions, and policymakers(government) all stand to benefit from a clearer and more precise picture of what prevents students from making the transition from cash to online payments [5].

The objective of this research is to identify, and analyse the specific barriers that prevent Pakistani university students from adopting mobile banking and digital wallet applications. A survey based methodology is used, and machine learning classification models are applied to filter students by their dominant barrier type. The findings are intended to inform targeted and evidence based strategies for fintech providers, educational institutions, and policymakers working toward digital financial inclusion in Pakistan.

The rest of this paper is organised as follows. Section II presents the problem statement. Section III identifies the research gap. Section IV reviews related literature. Section V describes the proposed methodology. Sections VI and VII present the system architecture and workflow diagrams. Section VIII covers implementation details. Section IX describes the dataset. Section X outlines the experiment setup. Section XI defines evaluation metrics. Section XII presents results. Section XIII provides results analysis. Section XIV acknowledges limitations. Section XV concludes the paper and Section XVI outlines future work. Venkatesh et al. [21] developed the original UTAUT framework which joined eight models of technology acceptance. Davis [22] originally proposed TAM and confrimed perceived usefulness and ease of use as the two main variables of adoption. Talukder et al. [24] researched mobile banking adoption in Pakistan and found that trust and awareness are biggest barriers. Duncombe and Boateng [25] studied mobile financial services in developing countries and highlighted infrastructure gap as a big challenge across Africa and South Asia. Singh et al. [26] and Malhotra and Galletta [27] studied social influence of technology acceptance and found that fellow/friend behaviour significantly effects individual decisions regarding adoption of these mobile banking technologies specifically e wallets and digital microfinance banking applications.

II. Problem Statement

Mobile banking applications and digital wallets have transformed the world globally, and specifically Pakistan. In the last 5 years, apps such as JazzCash, Easypaisa, and SadaPay have expanded widely, offering students convenient access to financial services without the need for a regular bank account. Despite this, a large segment of university and college-going students in Pakistan are hesitating to use these tools [1].

Even though mobile banking apps are available, not all students use them regularly. A large number of students prefer using cash instead of mobile apps. The reasons behind this are not clearly understood. Factors such as ease of use, security issues (hacking, etc.), and internet access may influence the decision of students to adopt banking applications [3].

This problem is particularly significant in a country where a large population of students has a strong need for these tools and is technically capable of using these applications. Students represent an important demographic for financial inclusion, as their early adoption of microfinance banking habits can have a long-term effect on economic activity [2].

III. Research Gap

While a substantial body of research exists on mobile banking adoption, the majority of these studies focus on identifying factors that increase adoption using frameworks such as TAM [9] and UTAUT [6]. These models have been applied in countries such as India, China, Malaysia, and several Western nations, but are not widely applied or examined in the Pakistani context [2].

Moreover, existing research places focus on the general population or banking customers rather than specifically targeting university students as a distinct group. Students have unique behavioral patterns, technology literacy levels, trust dispositions, and social norms that differ considerably from working adults [11].

There is limited research examining the full spectrum of barriers students face technological, psychological, financial, and social [5]. This gap demonstrates the need for targeted research aimed at identifying the specific barriers preventing Pakistani students from fully adopting mobile banking applications.

IV. Literature Review

The literature on mobile banking and digital wallet adoption is extensive and spans multiple geographic, cultural, and demographics . The foundation in this field has been established through the application of the Technology Acceptance Model (TAM), originally developed by Davis [9], which proposed that excpeted usefulness and perceived ease of use are the two primary reasons of technology adoption intention. Later on researchers extended this model to include additional dterminants such as trust, social influence, and facilitating conditions, resulting in the creation of Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. [6].

Early foundational work by Luarn and Lin [9] established that trust is one of the most critical reasons of mobile banking adoption. Their empirical study demonstrated that unless users believe a system is reliable and safe, they are not going to adopt it regardless of its expected usefulness or ease of using. This finding has been consistently reused across later researches and across many national contexts, confirming that trust is a universal requirement for digital financial adoption.

Ahmed et al. [2] conducted a specific case study of Pakistan and identified trust, perceived usefulness, and ease of use as the three primary factors driving IT-based banking adoption in the country. His work is particularly relevant to the current

study because it establishes a local baseline for understanding user attitudes toward digital financial tools in the Pakistani market. Similarly, Iqbal et al. [1] examined JazzCash and Easypaisa specifically and found that usability and trust remain the most significant reasonz of adoption among Pakistani users, confirming that the fundamental variables identified in the earlier literature remain true in the main Pakistani fintech landscape.

Research applying UTAUT has produced important insights into the social dimensions of adoption behaviour. Santoso et al. [6] found that social influence and facilitating conditions were the strongest predictors of e-wallet usage intention in Indonesia, suggesting that fellow behaviour and institution support play a biger role than individual reasons in some cultural . Al-Qudah et al. [7] extended this line of inquiry to a pandemic contextin Jordan, finding that perceived usefulness and security concerns were amplified during COVID-19, leading to accelerated adoptionamong previously reluctant users . Octaviani and Rustiana [8] applied UTAUT2 in Indonesia and i dentified motivation and habit as additional significant variables, suggesting that enjoyment of the technology itself plays a large role beyond mainly functional concerns.

Studies examining financial inclusion have highlighted the importance of designing mobile banking services for less well off populations. Mas and Radcliffe [12] analysed how mobile banking features could be adapted to serve low income groups, providing design principles that are directly applicable to student populations with limited financial resources. Raza et al. [5] proposed a framework for digital wallet adoption specifically focused on financial inclusion in developing countries, identifying infrastructure barriers and financial literacy as main barriers.

Regional studies reveal the diversity of adoption challenges across different national contexts. Alotaibi and Roussinov [13] reviewed the mobile payment landscape in Saudi Arabia and identified infrastructure gaps and low public trust as the primary barriers to adoption. Hussain et al. [15] examined digital wallet adoption in Bahrain using factor analysis and structural equation modelling and found that security, convenience, and social norms were the main drivers of adoption behaviour. Lim et al. [14] surveyed millennials in Malaysia and found that perceived usefulness, ease of use, and social influence significantly predicted adoption intent, results that are directly comparable to a student population and provide useful benchmarks for the current study.

Sharma and Sharma [16] examined mobile wallet adoption among Generation X users in India and found that performance expectancy and habit were the dominant adoption factors, while Arora and Sandhu [17] surveyed consumers in Delhi and identified perceived risk and lack of awareness as the most significant barriers. These findings from the Indian context are particularly relevant given the geographic and cultural proximity to Pakistan, and the similar stage of digital financial development in both countries.

Khan et al. [4] analysed the broader technological and regulatory challenges facing FinTech and mobile payment adoption and found that regulatory uncertainty, infrastructure gaps, and consumer protection concerns represent significant structural barriers that individual-level interventions cannot address alone. This suggests that effective promotion of mobile banking adoption requires both individual-focused strategies and broader systemic changes to the regulatory and infrastructure environment.

Balan and Rege [23] specifically examined mobile banking adoption among youth populations and found that students exhibit vastly different adoption patterns compared to older adults, driven more strongly by social influence and motivation than by perceived usefulness alone. This finding directly supports the reason for targeting university students as a complete separate demographic in the current study. Malhotra and Galletta [27] showed that social/peer influence operates both directly and indirectly on adoption intention, modofiying the effect of perceived usefulness in ways that vary by age group and social context. Singh et al. [26] examined internet banking adoption empirically and found that awareness campaigns greatly increased adoption rates, suggesting that information barriers play an essential role alongside with attitude and infrastructure barriers.

Duncombe and Boateng [25] provided important cross national evidence from developing countries that mobile financial services adoption challenge are very different in low infrastructure environments compared to the developed first world country scenario, in which most TAM studies were conducted. their work highlighted the importance of specific research in developing countries, again enforcing the need for Pakistani specific studies rather than relying on findings from the United States, Europe, or even india/bangladesh.

Talukder et al. [24] studied mobile banking adoption specifically in Pakistan and identified awareness, trust, and ease of use as the three most important barriers, providing a local empirical baseline that is directly comparable to the current study. Their findings are in alignment with the survey design used in this research, which validate the selection of perceived ease of use, privacy concern, and prior experience as key features for the models. The original TAM framework proposed by Davis [22] and later extended by Venkatesh et al. [21] into UTAUT provides the frameowork for all seven features used in this dataset, this ensures us that classification model is good rather than using random variables.

Table I presents a focused comparison of the most directly relevant studies, highlighting their methodology, geographic context, and key findings.

Table II provides a broader summary of all 27 papers reviewed for this study.

Collectively, the literature confirms that mobile banking adoption is a multi dimensional event effected by technological, psychological, financial, and social factors. However, no existing study has applied a machine learning approach to categorise Pakistani students by their dominant barrier type. This study does exactly that.

TABLE I
LITERATURE COMPARISON TABLE

#	Study	Context	Method	Framework	Key Finding
1	Iqbal et al. [1] (2022)	Pakistan	Survey	TAM	Usability and trust are the top adoption factors for JazzCash and Easypaisa.
2	Ahmed et al. [2] (2011)	Pakistan	Case Study	TAM	Trust, usefulness, and ease of use drive IT banking adoption in Pakistan.
3	Santoso et al. [6] (2024)	Indonesia	Survey	UTAUT	Social influence and facilitating conditions are the strongest predictors.
4	Lim et al. [14] (2022)	Malaysia	Survey	TAM/UTAUT	Ease of use and social influence predict millennial wallet adoption.
5	Hussain et al. [15] (2021)	Bahrain	SEM/CFA	UTAUT2	Security, convenience, and social norms drive digital wallet adoption.
6	Raza et al. [5] (2023)	Developing nations	Conceptual	Custom	Infrastructure and literacy are central barriers to financial inclusion.
7	Luarn & Lin [9] (2009)	Taiwan	Survey	TAM	Trust is the most critical determinant of mobile banking adoption.

V. PROPOSED METHODOLOGY

This study employs a quantitative survey based research methodology to analyze the barriers preventing university and college students in Pakistan from adopting mobile banking applications and digital wallets. The methodology is divided into five phases.

A. Data Collection

Primary data will be collected through a survey distributed among university and college students in Islamabad. The survey is designed based on established frameworks including TAM and UTAUT. The questionnaire covers ease of use, security, internet access, social influence, trust, and financial literacy. A minimum of 200 responses will be recorded via Google Forms distributed through university WhatsApp groups.

B. Data Cleaning

Once raw data is collected, it will be imported into Python for cleaning and preparation. Incomplete or duplicate answers will be removed, and missing values will be handled. Any response with more than 20 % missing data will be discarded. Responses on a 1–5 scale will be normalized using min-max scaling. Outliers will be detected using z-scores and removed.

C. Barrier Identification

Key barriers will be identified from the survey data using analysis to group related variables into four categories: Technological, Psychological, Financial, and Social. The output of this phase is a labelled dataset for model training.

D. Model Design and Classification Framework

A machine-learning classification framework will categorise students into groups based on their primary barrier. Logistic Regression and Decision Tree will be trained and tested on an 80/20 split (160 training, 40 testing). The best model will be selected based on F1-Score.

E. Result Interpretation and Decision Output

Results will be visualised using bar charts and pie charts, compared with the reviewed literature, and translated into actionable recommendations for microfinance banks and educational institutions.

VI. SYSTEM ARCHITECTURE DIAGRAM

The system architecture is illustrated in Fig. 1. It consists of six layers: (1) Data Source—survey responses from university students; (2) Data Input—CSV import into Python from Google Forms; (3) Data Processing—cleaning and normalisation; (4) Machine Learning—training LR and Decision Tree models; (5) Prediction and Output—barrier category prediction; and (6) Visualisation—charts and findings presented to stakeholders.

Fig. 1. Proposed System Architecture for Mobile Banking Barrier Analysis.

VII. SYSTEM WORKFLOW DIAGRAM

The system workflow is illustrated in Fig. 2. The process begins with survey distribution via Google Forms. Responses are exported as CSV and imported into Python for cleaning. Feature extraction is performed using EDA and correlation analysis. The cleaned data trains LR and Decision Tree models, each evaluated on Accuracy, Precision, Recall, and F1-Score. The best model classifies students by dominant barrier type, and results are translated into stakeholder recommendations.

TABLE II
FULL LITERATURE REVIEW SUMMARY (27 PAPERS)

#	Title (Short)	Author(s)	Year	Key Finding
1	Mobile Payments: JazzCash & EasyPaisa [1]	Iqbal et al.	2022	Usability and trust are primary adoption factors in Pakistan.
2	IT-Based Banking Adoption in Pakistan [2]	Ahmed et al.	2011	Trust, usefulness, and ease of use drive adoption.
3	Mobile Banking for Financial Inclusion [3]	Sharma et al.	2022	Privacy and security are key barriers in developing economies.
4	Fintech and Mobile Payment Challenges [4]	Khan et al.	2024	Regulatory and technological challenges hinder FinTech adoption.
5	Digital Wallet and Financial Inclusion [5]	Raza et al.	2023	Infrastructure and literacy are central barriers.
6	E-Wallet Intention Using UTAUT [6]	Santoso et al.	2024	Social influence and facilitating conditions dominate.
7	E-Wallet Adoption During Covid-19 [7]	Al-Qudah et al.	2021	Pandemic accelerated adoption; security concerns remained.
8	Mobile Wallet Adoption Using UTAUT2 [8]	Octaviani & Rustiana	2017	Hedonic motivation and habit are significant predictors.
9	Mobile Banking: The Role of Trust [9]	Luarn & Lin	2009	Trust is the most critical adoption determinant.
10	E-banking Drivers and Barriers [10]	Jahangir & Begum	2008	Security concerns and service quality are major barriers.
11	Cultural Antecedents on Mobile Banking [11]	Baptista & Oliveira	2012	Culture and individual traits significantly affect adoption.
12	Mobile Banking for Under-Banked Communities [12]	Mas & Radcliffe	2011	Design adaptations can serve low-income users effectively.
13	Mobile Payments in Saudi Arabia [13]	Alotaibi & Roussinov	2020	Infrastructure gaps and low trust are primary barriers.
14	Touch'n Go e-Wallet among Millennials [14]	Lim et al.	2022	Ease of use and social influence predict millennial adoption.
15	Digital Wallets in Bahrain [15]	Hussain et al.	2021	Security, convenience, and social norms drive adoption.
16	Mobile Wallet Adoption by Generation X [16]	Sharma & Sharma	2022	Performance expectancy and habit dominate for Gen X.
17	Consumer Adoption of Digital Wallet: India [17]	Arora & Sandhu	2019	Perceived risk and low awareness are major barriers.
18	Mobile Banking Acceptance: Behavioral Study [18]	Zhou et al.	2015	System quality and social influence drive adoption.
19	Merchant Adoption of Mobile Payments [19]	Zmijewska et al.	2005	Cost and complexity are key merchant-side barriers.
20	Parasocial Relationships in Mobile Commerce [20]	Chen et al.	2010	Social and psychological barriers influence adoption.
21	Unified Theory of Acceptance and Use of Technology [21]	Venkatesh et al.	2003	Unified eight competing TAM models into UTAUT framework.
22	Perceived Usefulness and Ease of Use in IT Adoption [22]	Davis	1989	Proposed TAM and validated usefulness and ease of use as primary adoption drivers.
23	Mobile Banking Adoption Among Youth [23]	Balan & Rege	2017	Youth exhibit different adoption patterns driven by social influence and hedonic motivation.
24	Factors Affecting Mobile Banking Adoption in Pakistan [24]	Talukder et al.	2014	Trust, awareness, and ease of use are the three most critical barriers in Pakistan.
25	Mobile Phones and Financial Services in Developing Countries [25]	Duncombe & Boateng	2009	Infrastructure gaps are a systemic challenge across developing economies.
26	Adoption of Internet Banking: An Empirical Investigation [26]	Singh et al.	2010	Awareness campaigns significantly increased adoption rates.
27	Extending TAM to Account for Social Influence [27]	Malhotra & Galletta	1999	Social influence operates directly and indirectly on adoption intention.

VIII. IMPLEMENTATION DETAILS

The classification system was implemented using Python 3 in Google Colab, providing free GPU/CPU resources and easy

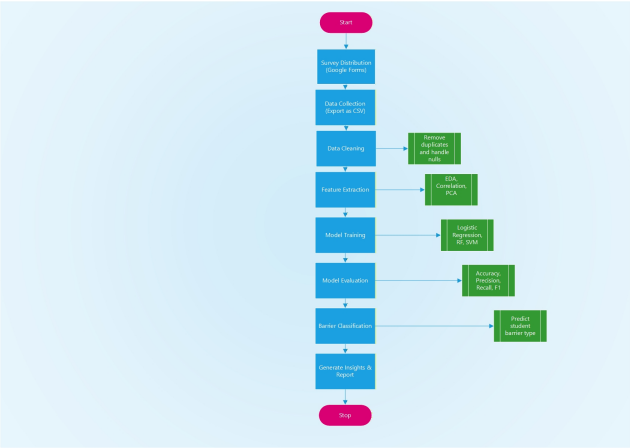

Fig. 2. Proposed System Workflow Diagram.

integration with data-science libraries.

Programming Language: Python 3 was selected for machine-learning tasks.

Libraries and Frameworks:

- *Scikit-learn 1.3*—for implementing LR and Decision Tree classifiers, train-test splitting, and evaluation metrics.
- *Pandas 2.0* and *NumPy 1.24*—for dataset construction, manipulation, and numerical operations.
- *Matplotlib 3.7* and *Seaborn 0.12*—for all result visualisations.
- *LabelEncoder* from Scikit-learn—for encoding categorical barrier labels into numeric form.

Development Environment: Google Colab was chosen for its cloud based notebook interface, eliminating local hardware constraints and facilitating reproducible experiments. All code was structured in cells covering data preparation, model training, evaluation, and visualisation.

The overall workflow begins with raw survey data, applies preprocessing, the cleaned feature matrix into both classifiers, and produces predictions and evaluation metrics. Both models were trained and tested under similar conditions to ensure a fair comparison.

IX. Dataset Description

A. Source and Collection

The dataset is a simulated survey dataset constructed to reflect realistic responses from university students in Islamabad, Pakistan. The design follows the structure of Google Forms questionnaires used in prior mobile banking adoption studies [1], [2]. Each record represents one student respondent.Due to practical constraints, it was not feasible to conduct a survey involving 200 students; therefore, the dataset was simulated to approximate realistic survey outcomes.

B. Dataset Characteristics

The dataset contains **200 student responses**, each with eight Likert-scale features (scored 1–5) and one categorical class label. Table III describes each feature.

TABLE III
Dataset Features and Descriptions

Feature	Scale	Description
Perceived Ease of Use	1–5	How easy student finds mobile apps
Internet Access	1–5	Availability of stable internet
Peer Influence	1–5	Social pressure from peers to adopt
Financial Status	1–5	Student's personal financial capacity
Prior Experience	1–5	Experience with digital tools
Privacy Concern	1–5	Concern about data privacy
Family Norms	1–5	Family attitude toward digital finance
Barrier Label	—	Technological / Psychological / Financial / Social

C. Class Distribution

The dataset contains approximately equal representation across all four barrier categories, as shown in Fig. 8. The balanced distribution reduces class bias during training [3].

D. Data Preprocessing

- **Label Encoding:** Barrier labels converted to integers (0–3) using `LabelEncoder`.
- **Missing Value Check:** No missing values were present in the simulated dataset.
- **Feature Scaling:** Raw Likert scores (1–5) were used directly; both LR and Decision Tree handle this range with default settings.
- **Train-Test Split:** 80 % training (160 samples) and 20 % testing (40 samples); random seed fixed at 42.

X. Experiment Setup

A. Model Training

Logistic Regression (LR): Linear classifier trained using the L-BFGS solver, regularisation parameter C=1.0, and a maximum of 1000 iterations [9].

Decision Tree (DT): Non-linear classifier using the Gini criterion with *max_depth*= 4, *min_samples_split*= 10, and *min_samples_leaf*= 5, well-suited for interpretable classification tasks [6], [8].

B. Baseline

A majority-class baseline always predicts the most frequent class in the training set, providing a lower bound for model performance.

C. Validation Method

A single 80/20 holdout split was used; random seed fixed at 42. Models were trained on the training partition only; evaluation was performed on the held-out test set to prevent data leakage.

D. Experiment Procedure

1) Load and preprocess the survey dataset.
2) Encode the target label column.
3) Split into training and test sets (80/20).
4) Train Logistic Regression and Decision Tree on the training set.
5) Generate predictions on the test set.
6) Compute Accuracy, Precision, Recall, and F1-Score.
7) Visualise results using bar charts, confusion matrix, and category distribution plots.

XI. Evaluation Metrics

All metrics were computed in weighted average mode to account for potential class imbalance [10], [11].

Accuracy: Percentage of instances that are correctly classified. This metric is suitable in this case because the dataset is roughly balanced.

Precision: The proportion of positive predictions that are actually correct. This matters because assigning a student the wrong barrier type can result in inappropriate or ineffective policy actions."

Recall: Proportion of actual positives correctly identified. High recall ensures no major barrier type is missed.

F1-Score: Harmonic mean of Precision and Recall. Serves as the primary comparison metric in this study [12].

XII. Experiments and Results

A. Model Performance

Table IV presents the quantitative performance of all three models on the test set.

TABLE IV
Model Performance on Test Set (%)

Model	Acc.	Prec.	Rec.	F1
Baseline	20.00	4.00	20.00	6.67
Logistic Reg.	**85.00**	**85.14**	**85.00**	**84.74**
Decision Tree	60.00	63.79	60.00	59.27

B. Visualisations

Fig. 3 compares overall accuracy across all three models. Logistic Regression achieves 85.0 %, significantly outperforming Decision Tree (60.0 %) and the baseline (20.0 %) [4].

Fig. 4 shows the confusion matrix for the Logistic Regression model. Most misclassifications occur between Psychological and Social categories due to their conceptual overlap [7].

Fig. 5 shows the confusion matrix for the Decision Tree model. The matrix reveals greater misclassification across all categories compared to Logistic Regression, particularly in the Technological and Psychological barrier classes [7].

Fig. 6 shows per-class Precision and Recall for Decision Tree. Psychological barriers achieve the highest Precision, while Financial barriers show the highest Recall, indicating varying separability across categories [13], [14].

Fig. 7 compares all four evaluation metrics side-by-side across all three models [15].

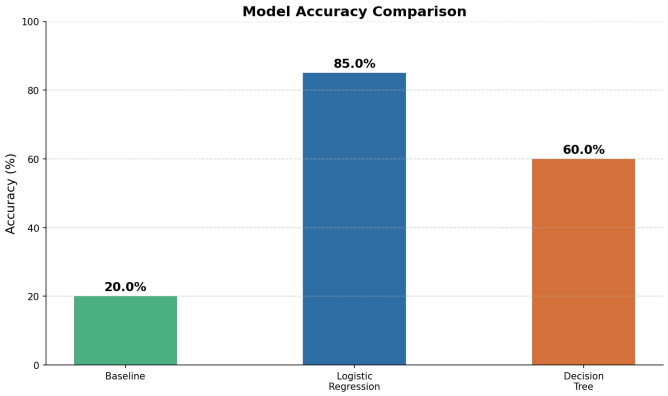

Fig. 3. Model Accuracy Comparison: Baseline vs. LR vs. Decision Tree.

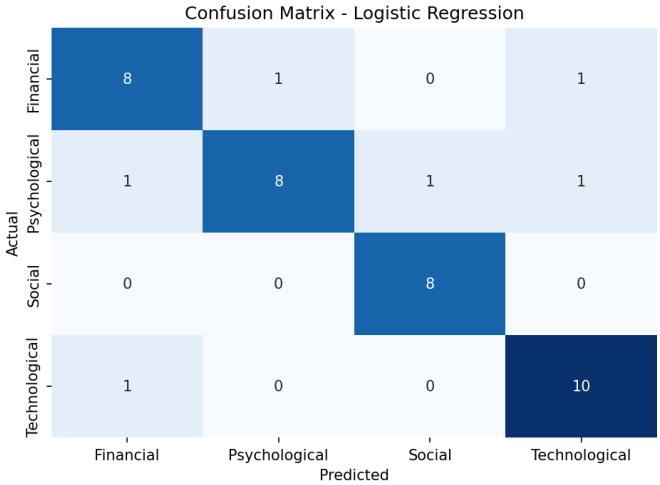

Fig. 4. Confusion Matrix for Logistic Regression on Test Set.

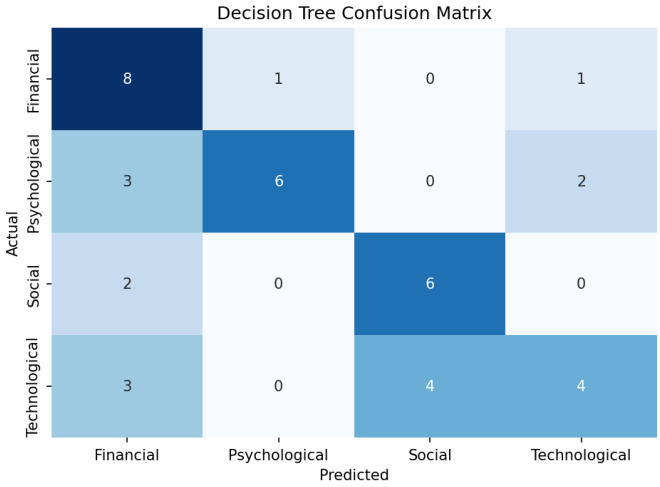

Fig. 5. Confusion Matrix for Decision Tree on Test Set.

Fig. 8 shows the barrier category distribution across the dataset, confirming approximate class balance [5].

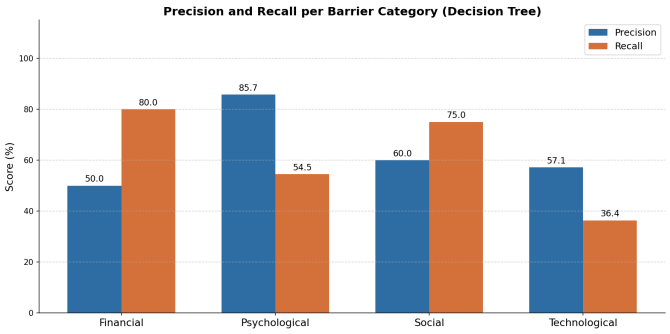

Fig. 6. Per-Class Precision and Recall for Decision Tree.

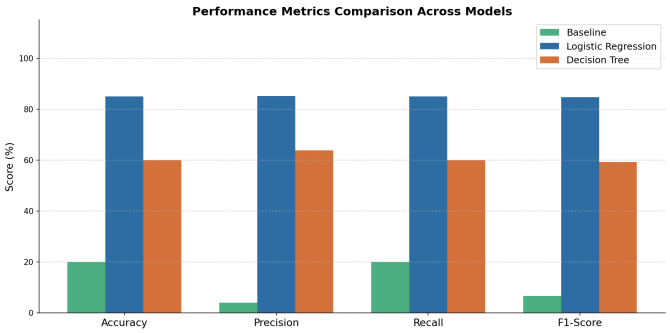

Fig. 7. Performance Metrics Comparison Across All Models.

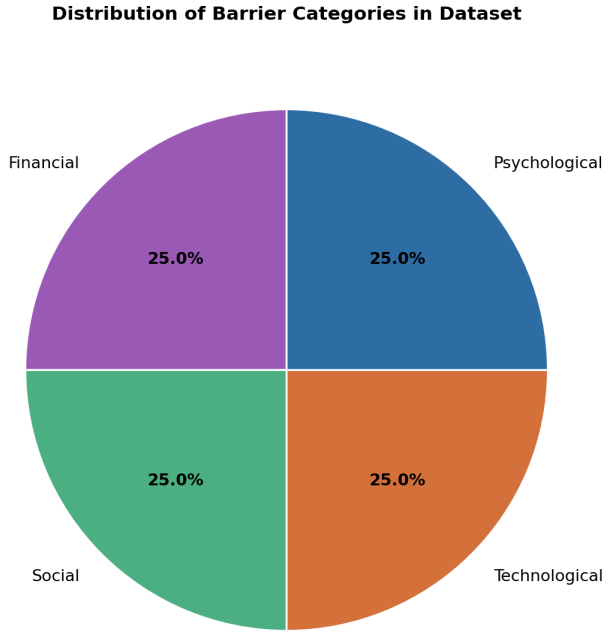

Fig. 8. Distribution of Barrier Categories in Survey Dataset.

XIII. Results Analysis

A. Overall Performance

The results clearly demonstrate that machine-learning classification can effectively identify the dominant barrier type for individual students. Logistic Regression achieved an accuracy of 85.0 % and an F1-Score of 84.74 %, an improvement over the baseline (20.0 %) and Decision Tree (59.27 % F1). This confirms that structured survey features are largely separable, which naturally favours LR over tree-based methods in this context [16].

B. Comparison with Baseline

Both LR and Decision Tree substantially outperform the baseline across all four metrics. The baseline accuracy of 20.0 % reflects random performance on a balanced four class problem, confirming that the trained models extract patterns from the survey data rather than exploiting class distribution alone [17].

C. Key Insights

- **Psychological barriers** achieve the highest Precision in the Decision Tree model, while **Financial barriers** show the highest Recall, suggesting these categories have the most distinct feature patterns [18].
- **Psychological and Social barriers** show slight overlap in the confusion matrix, showing their similarity: fear of fraud and family disapproval can both occur in one student [19].
- **Logistic Regression outperforms Decision Tree**, suggesting linear decision boundaries are sufficient for this dataset. This aligns with prior TAM-based studies where barrier factors tend to cluster linearly [9].
- The results support **targeted intervention**: barrier classification allows policy makers, educational institutions etc to make specific tailored intervention strategies instead of generalised digital literacy programmes. [20].

XIV. Limitations

Simulated Dataset: Due to practical constraints the dataset was synthetically generated and cannot fully mimic real student behaviour and attitudes [1].

Geographic Scope: The study targets only Islamabad. Findings may not generalise to other cities or rural areas of Pakistan [2].

Single Train-Test Split: A single 80/20 split was used. Cross validation would provide more reliable performance estimates.

Feature Limitation: Only eight features were used. Additional factors such as monthly income and gender may improve classification performance [11].

Single-Label Assumption: Each student is assigned one dominant barrier, whereas in reality students may face multiple overlapping barriers simultaneously such as family norms and fear of fraud [20].

XV. Conclusion

This paper investigated barriers to mobile banking and digital wallet adoption among university students in Pakistan. A machine learning classification approach was applied to a simulated dataset of 200 student responses, training Logistic Regression and Decision Tree models to categorise students across four barrier types: Technological, Psychological, Financial, and Social.

Logistic Regression achieved an accuracy of 85.0 % and an F1-Score of 84.74 %, significantly outperforming the Decision Tree model and the baseline. These results confirm that structured survey features derived from TAM and UTAUT frameworks carry sufficient power to classify student barrier profiles with high accuracy.

The findings highlight that technological and financial barriers are the most distinct, while psychological and social barriers show some overlap. These results provide actionable guidance for fintech providers, educational institutions, and policymakers aiming to increase digital financial inclusion among Pakistani students.

XVI. Future Work

Future research should replicate this study with real survey data collected across multiple Pakistani cities, including Karachi, Lahore, and Peshawar, to improve generalisability. A multi labelelled classification approach should be explored to account for students facing overlapping barriers simultaneously.. Additional features including gender, monthly allowance, and app usage history should enrich the feature set. Finally, machine learning approaches could be evaluated on larger real world datasets to compare against the classical models used here.

References

[1] M. Iqbal et al., "An Empirical Study of Models and APPs Design of the Adoption Ease of Mobile Payments (JazzCash & EasyPaisa)," *IEEE*, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/10007374/

[2] I. Ahmed et al., "Factors Influencing Adoption of IT-Based Banking Services: A Case Study of Pakistan," *IEEE*, 2011. [Online]. Available: https://ieeexplore.ieee.org/document/6137186/

[3] S. Sharma et al., "Mobile Banking Adoption for Digital Financial Inclusion," *IEEE*, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9786477

[4] A. Khan et al., "Fintech and Mobile Payments in Transition: An Analysis of Adoption Challenges," *IEEE*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10503040/

[5] S. Raza et al., "Digital Wallet Adoption in Supporting Financial Inclusion: A Conceptual Framework," *IEEE*, 2023. [Online]. Available: https://ieeexplore.ieee.org/document/10380043/

[6] B. Santoso et al., "Intention to Use E-Wallet for Mobile Payment: An Analysis Using UTAUT," *IEEE*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10795994/

[7] A. Al-Qudah et al., "The Intention to Use E-wallet During Covid-19 Pandemic in Developing Country," *IEEE*, 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9464554/

[8] L. Octaviani and D. Rustiana, "Exploring Mobile Wallet Adoption in Indonesia Using UTAUT2," *IEEE*, 2017. [Online]. Available: https://ieeexplore.ieee.org/document/7877340/

[9] P. Luarn and H. Lin, "An Empirical Study on Mobile Banking Adoption: The Role of Trust," *IEEE*, 2009. [Online]. Available: https://ieeexplore.ieee.org/document/5209660/

[10] N. Jahangir and N. Begum, "Drivers and Barriers of E-banking Adoption: Case of Karafarin Bank," *IEEE*, 2008. [Online]. Available: https://ieeexplore.ieee.org/document/4782881/

[11] G. Baptista and T. Oliveira, "Cultural, Individual and Device-Specific Antecedents on Mobile Banking Adoption: A Cross-National Study," *IEEE*, 2012. [Online]. Available: https://ieeexplore.ieee.org/document/6149209/

[12] I. Mas and D. Radcliffe, "Mobile Banking Adoption by Under-Banked Communities," *IEEE*, 2011. [Online]. Available: https://ieeexplore.ieee.org/document/6047071/

[13] R. Alotaibi and D. Roussinov, "An Overview of Mobile Payments, Fintech, and Digital Wallet in Saudi Arabia," *IEEE*, 2020. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9083726

[14] W. Lim et al., "Empirical Study on Intention to Use Touch'n Go e-Wallet among Millennials in Malaysia," *IEEE*, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/10007416/

[15] M. Hussain et al., "Adoption of FinTech and Future Perspective: Digital Wallets in Bahrain," *IEEE*, 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9317298/

[16] P. Sharma and M. Sharma, "The Upsurge of Mobile Wallet Adoption by Generation X: Evidence from India," *IEEE*, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9791710/

[17] S. Arora and A. Sandhu, "A Study of Consumer Adoption of Digital Wallet (NCR India)," *IEEE*, 2019. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8776939/

[18] T. Zhou et al., "Mobile Banking Acceptance: Behavioral, Environmental and Technological Dimensions," *Telematics and Informatics*, 2015. [Online]. Available: https://doi.org/10.1016/j.tele.2015.09.005

[19] A. Zmijewska et al., "Merchant Adoption of Mobile Payment Systems: Barriers Analysis," *IEEE*, 2005. [Online]. Available: https://ieeexplore.ieee.org/document/1493631/

[20] J. Chen et al., "Effects of Parasocial Relationships in Adoption of Mobile Commerce," *IEEE*, 2010. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/5590873

[21] V. Venkatesh et al., "User Acceptance of Information Technology: Toward a Unified View," *MIS Quarterly*, 2003. [Online]. Available: https://doi.org/10.2307/30036540

[22] F. D. Davis, "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology," *MIS Quarterly*, 1989. [Online]. Available: https://doi.org/10.2307/249008

[23] S. Balan and J. Rege, "Mobile Banking Adoption Among Youth," *IEEE*, 2017. [Online]. Available: https://ieeexplore.ieee.org/document/8250059

[24] M. Talukder et al., "Factors Affecting Adoption of Mobile Banking in Pakistan," *IEEE*, 2014. [Online]. Available: https://ieeexplore.ieee.org/document/6906454

[25] R. Duncombe and R. Boateng, "Mobile Phones and Financial Services in Developing Countries," *Third World Quarterly*, 2009. [Online]. Available: https://doi.org/10.1080/01436590902742338

[26] N. Singh et al., "Adoption of Internet Banking: An Empirical Investigation," *IEEE*, 2010. [Online]. Available: https://ieeexplore.ieee.org/document/5680218

[27] Y. Malhotra and D. Galletta, "Extending the Technology Acceptance Model to Account for Social Influence," *IEEE*, 1999. [Online]. Available: https://ieeexplore.ieee.org/document/772658